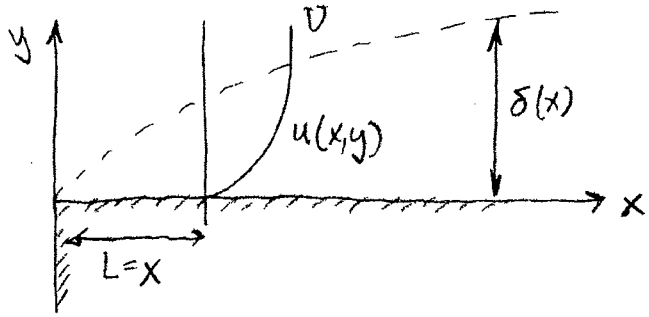
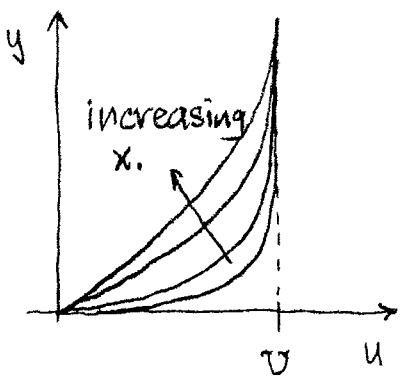
Similarity Solution to Blasius Boundary Layer:



Since x and y are semi infinite, there is no characteristic geometric scale.

Thus, we may expect a similarity solution. Indeed, the velocity profile maintains a constant shape:



In non-dimensional space, our equations were:

$$x^* = \frac{x}{L} \qquad u^* = \frac{u}{U}$$

$$y^* = \frac{y}{\delta} \qquad v^* = \frac{v}{V} = \frac{v}{U\delta/L}$$

Giving:

$$\frac{\partial u^*}{\partial x^*} + \frac{\partial v^*}{\partial y^*} = 0$$

$$u^*\frac{\partial u^*}{\partial x^*} + v^*\frac{\partial u^*}{\partial y^*} = \frac{1}{Re_L}\frac{L^2}{\delta^2}\frac{\partial^2 u^*}{\partial y^{*2}}$$

Both terms must be of order 1.

Thus, we deduce:

$$\frac{1}{Re_L}\frac{L^2}{\delta^2} \approx 1$$

$\downarrow$ solve for δ:

$$\delta = \sqrt{\frac{\nu L}{U}}$$

Substitute δ and $L = x$ into our scale definitions:

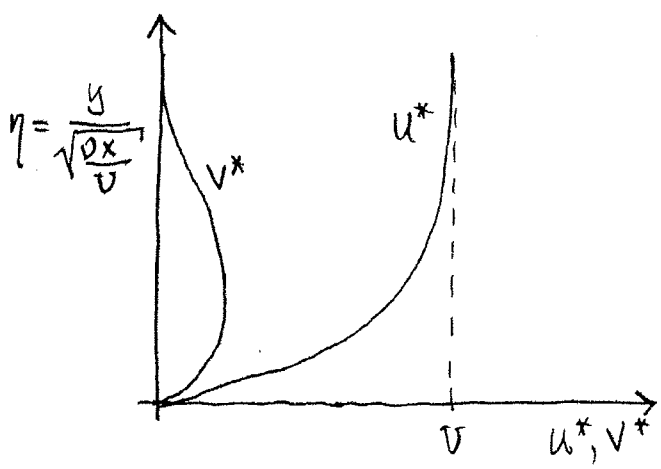
$$x^* = \frac{x}{x} = 1 \qquad u^* = \frac{u}{U}$$

$$\underbrace{y^* = \frac{y}{\delta} = \frac{y}{\sqrt{\nu x/U}}}_{} \qquad v^* = \frac{v}{\sqrt{\nu U/x}}$$

We have eliminated the dependence on x^*.

↳ y^* is our similarity variable

Our similarity solution will give:



And all data at any x and y will collapse to these curves.

Blasius Boundary Layer:

We assume a solution of the form:

$$\left.\begin{aligned} u &= U\,F(\eta) \\ v &= \sqrt{\frac{U\nu}{x}}\,G(\eta) \end{aligned}\right\} \quad \text{with } \eta = \frac{y}{\sqrt{\frac{\nu x}{U}}}$$

For the chain rule, we will need:

$$\frac{\partial \eta}{\partial x} = \frac{\partial}{\partial x}\left(y\sqrt{\frac{U}{\nu x}}\right) = -\frac{1}{2}\underbrace{y\sqrt{\frac{U}{\nu x}}}_{\eta}\frac{1}{x} = -\frac{\eta}{2x}$$

$$\frac{\partial \eta}{\partial y} = \frac{\partial}{\partial y}\left(y\sqrt{\frac{U}{\nu x}}\right) = \sqrt{\frac{U}{\nu x}}$$

We also need:

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x}\left(U F(\eta)\right) = U\frac{\partial F}{\partial \eta}\frac{\partial \eta}{\partial x} = -\frac{U\eta}{2x}\frac{\partial F}{\partial \eta}$$

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}\left(U F(\eta)\right) = U\frac{\partial F}{\partial \eta}\frac{\partial \eta}{\partial y} = U\sqrt{\frac{U}{\nu x}}\frac{\partial F}{\partial \eta}$$

$$\frac{\partial v}{\partial y} = \frac{\partial}{\partial y}\left(\sqrt{\frac{U\nu}{x}}\,G(\eta)\right) = \sqrt{\frac{U\nu}{x}}\frac{\partial G}{\partial \eta}\cdot\frac{\partial \eta}{\partial y} = \frac{U}{x}\frac{\partial G}{\partial \eta}$$

Boundary Conditions:

@ $y=0$: $\quad u(x,0)=0 \rightarrow F(0)=0$

$\qquad\qquad v(x,0)=0 \rightarrow G(0)=0$

Match : $\quad u(x,\infty)=U \rightarrow F(\infty)=1$

Conservation of Mass:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{\cancel{\nu}}{\cancel{x}}\left[-\frac{\eta}{2}\frac{\partial F}{\partial \eta} + \frac{\partial G}{\partial \eta}\right] = 0$$

$$\frac{dG}{d\eta} = \frac{\eta}{2}\frac{dF}{d\eta}$$

ODE since G, F are functions of η only.

To solve for G, introduce a new unknown function f:

$$F = \frac{df}{d\eta}$$

; f = slope of F

≡ definition of a streamline.

↳ f is the stream function.

Then, substituting and integrating:

$$\int \frac{dG}{d\eta}\, d\eta = \int \frac{\eta}{2}\frac{d^2 f}{d\eta^2}\, d\eta$$

integrate by parts: $\int u\,dv = uv - \int v\,du$

$u = \frac{\eta}{2}$ $\quad dv = \frac{d^2 f}{d\eta^2}$

$du = \frac{d\eta}{2}$ $\quad v = \frac{df}{d\eta}$

$$G = \frac{\eta}{2}\frac{df}{d\eta} - \int \frac{1}{2}\frac{df}{d\eta}\, d\eta + \cancel{C}$$

choose ∅: only affects the magnitude of the stream function.

$$G = \frac{1}{2}\left(\eta\frac{df}{d\eta} - f\right)$$

Conservation of Momentum:

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \nu\frac{\partial^2 u}{\partial y^2}$$

$$UF\left(-\frac{U\eta}{2x}\frac{\partial F}{\partial \eta}\right) + \sqrt{\frac{U\nu}{x}}\,G(\eta)\,U\sqrt{\frac{U}{\nu x}}\frac{\partial F}{\partial \eta} = \nu U\cdot\frac{U}{\nu x}\frac{\partial^2 F}{\partial \eta^2}$$

$$\frac{\cancel{U^2}}{\cancel{x}}\left[G\frac{dF}{d\eta} - \frac{\eta F}{2}\frac{dF}{d\eta} = \frac{d^2F}{d\eta^2}\right]$$ ← ODE since F, G functions of η only.

Substitute:

$$F = \frac{df}{d\eta} \quad \text{and} \quad G = \frac{1}{2}\left(\eta\frac{df}{d\eta} - f\right)$$

Giving:

$$\frac{1}{2}\left(\cancel{\eta\frac{df}{d\eta}} - f\right)\frac{d^2f}{d\eta^2} - \cancel{\frac{\eta}{2}\frac{df}{d\eta}\frac{d^2f}{d\eta^2}} = \frac{d^3f}{d\eta^3}$$

$$\underline{\underline{\frac{d^3f}{d\eta^3} + \frac{f}{2}\frac{d^2f}{d\eta^2} = 0}}$$

Boundary Conditions:

$$F(0) = 0 \rightarrow \frac{df}{d\eta} = 0 \text{ at } \eta = 0$$

$$G(0) = 0 \rightarrow f = 0 \text{ at } \eta = 0$$

$$F(\infty) = 1 \rightarrow \underline{\frac{df}{d\eta} = 1 \text{ at } \eta = \infty}$$

Solution:

Solve for f using 4th. order Runga-Kutta scheme.
Convert to system of ODE's:

$$\frac{f}{2}\frac{d^2f}{d\eta^2} + \frac{d^3f}{d\eta^3} = 0$$

Define:

$$F = \frac{df}{d\eta}$$

$$H = \frac{dF}{d\eta} = \frac{d^2f}{d\eta^2}$$

Then we have:

$$\frac{dH}{d\eta} = -\frac{f}{2}H$$

$$\frac{dF}{d\eta} = H$$

$$\frac{df}{d\eta} = F$$

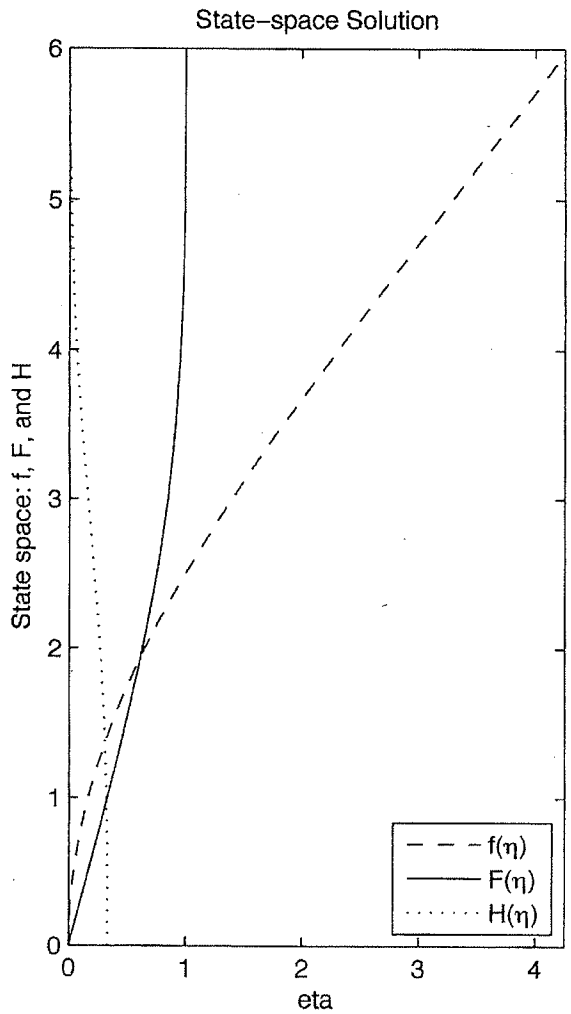
Subject to the conditions:

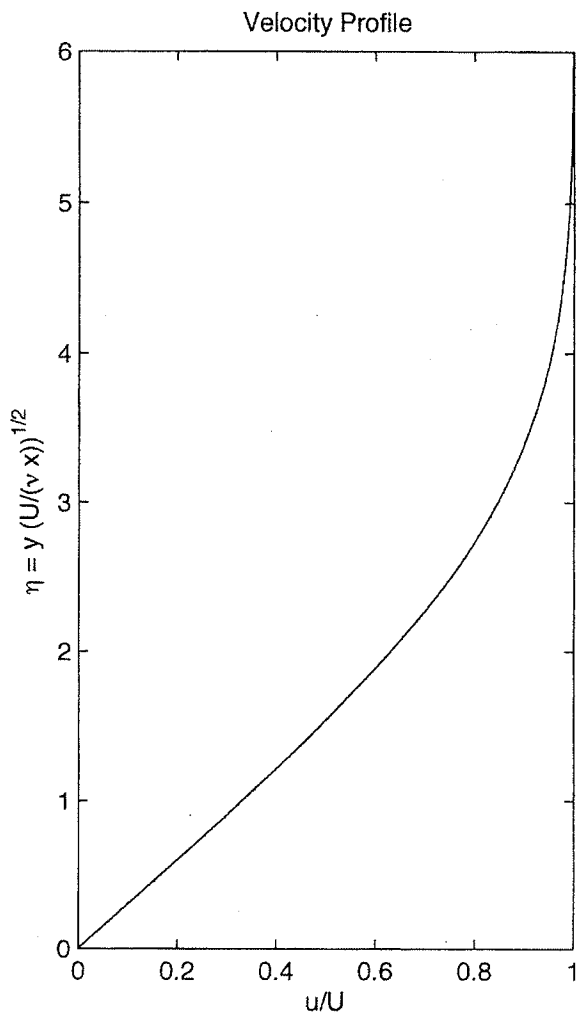
$$f(0) = 0$$

$$F(0) = 0$$

$$F(\infty) = 1$$

Solution is considered "Exact" since equation is an ODE and numerical method has very low error.


State−space Solution
State space: f, F, and H
eta
f(η)
F(η)
H(η)
0
1
2
3
4
5
6


Velocity Profile
η = y (U/(ν x))^1/2
u/U
0
0.2
0.4
0.6
0.8
1
2
3
4
5
6